

# Cross-Subject Human Activity Recognition on 1 Hz Wrist Accelerometry: Calibrated GBDT Stacking with Orientation Pseudo-Gyroscope and Test-Prior Adaptation

Data Mining Final Project, 2026 Spring

吳忠賢 Michael Kurniawan Soegeng  
Student ID 314540066

**Abstract**—We tackle the NYCU Data Mining 2026 Assignment 3 Kaggle competition: six-class human activity recognition (HAR) from 1 Hz aggregated tri-axial wrist accelerometry, evaluated by macro-F1 under a *cross-subject* (disjoint train/test users) split. Our final system is a calibrated gradient-boosted-tree (GBDT) stack blended with a hierarchical pipeline, augmented by two ideas tailored to the data physics: (i) an *orientation pseudo-gyroscope*—because gravity is not removed, the per-second mean trace recovers wrist orientation whose derivative approximates the missing gyroscope—and (ii) a label-free *test-prior correction* that adapts predictions to the measured test class distribution. The system reaches public-leaderboard macro-F1 0.8234, up from a 0.7473 single-model baseline. Our code is available at GitHub: <https://github.com/Ashurali/DM2026-Assignment-3-MKS> (see README.md for run instructions; the final result reproduces with `python scripts/reproduce_final.py`).

## I. PROJECT SUMMARY

**Goal.** Classify each 300-second wrist-accelerometer recording into one of six activities. The metric is macro-F1, so the rare classes matter as much as the common ones, and the train/test users are *completely disjoint*, making cross-subject generalization the central challenge.

**High-level idea.** Rather than rely on a single model, we (1) engineer a broad time-series feature catalog and stack diverse base learners with a GBDT meta-model [1]; (2) blend this flat stack with a coarse→fine hierarchical pipeline; (3) calibrate per class and tune per-class decision thresholds to recover minority recall; and finally (4) exploit two properties of *this* dataset—gravity-laden signals (an orientation “pseudo-gyroscope”) and a shifted test class prior (label-shift adaptation [2]). Each stage is validated under strict GroupKFold by user.

**Related work.** Kaggle-style competitions are frequently won by careful feature engineering and model ensembling [3], [4]. Human activity recognition from inertial sensors is classically tackled either by hand-crafted statistical/spectral features fed to a classifier or by end-to-end deep networks on the raw stream; on small, tabular-friendly problems gradient-boosted trees [1] remain highly competitive, which motivates our feature-plus-GBDT backbone. For time series we additionally

draw on canonical feature sets (`catch22` [5]), convolutional transforms (MiniRocket [6]), and deep architectures (InceptionTime [7]). A recurring theme in cross-subject HAR is that models latch onto user-specific idiosyncrasies; our strict per-user cross-validation, feature pruning, and—most importantly—test-time prior adaptation [2] together with isotonic probability calibration [8] directly target that distribution shift. The dataset originates from a public ADL corpus of wrist-placed accelerometers [9].

## II. PROPOSED METHOD

### A. Overview

Figure 1 shows the full pipeline. Two base predictors feed a blend that is calibrated, then adapted at test time and thresholded. We describe each component below. The design follows one principle: on a small, imbalanced, cross-subject problem, *robust decoding beats raw capacity*. Stacking heterogeneous base models supplies diversity, disciplined post-processing (calibration, threshold tuning, prior adaptation) converts that diversity into macro-F1, and the two physics-aware additions inject information that the 1 Hz signal hides rather than merely re-fitting the training distribution.

### B. Feature representation and temporal modeling

Each file is a sequence  $X \in \mathbb{R}^{6 \times 300}$  (per-second mean $_{x,y,z}$  and std $_{x,y,z}$ ) with a single label. We summarize temporal structure into an 11-family, 271-dimensional catalog: per-channel statistics, FFT band energies (rhythmicity), autocorrelation (periodicity), sub-window pooling (phase), jerk (first difference), zero-crossing rates, magnitude, gravity, and cross-axis terms; we further add the `catch22` canonical descriptors [5]. To capture dependencies the summaries miss, we also train sequence models on the raw  $6 \times 300$  input (CNN-BiLSTM, Transformer, MiniRocket [6], InceptionTime [7]) and stack their out-of-fold (OOF) probabilities.

**Orientation pseudo-gyroscope.** Because gravity is *not* removed and the device is wrist-worn, the per-second mean vector  $\mathbf{g}_t = (\text{mean}_x, \text{mean}_y, \text{mean}_z)_t$  traces the wrist’s *orientation trajectory*. We derive pitch  $\theta_t = \text{atan2}(g_{x,t}, \sqrt{g_{y,t}^2 + g_{z,t}^2})$ , roll

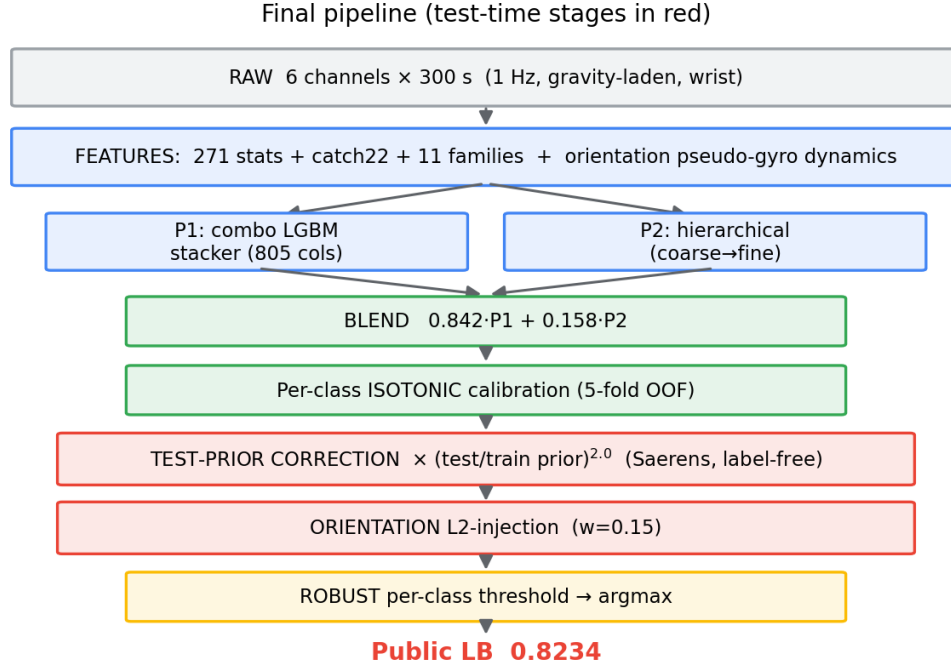


Fig. 1. Final pipeline. Feature extraction feeds a flat GBDT stack (P1) and a hierarchical model (P2); their blend is isotonic-calibrated, adapted to the measured test prior, refined on the bottleneck class by the orientation pseudo-gyroscope source, and decoded by per-class thresholds. Test-time stages are shown in red.

$\phi_t = \text{atan2}(g_{y,t}, g_{z,t})$ , and inclination  $\psi_t = \arccos(g_{z,t}/\|\mathbf{g}_t\|)$ , and treat their per-second differences  $(\Delta\theta_t, \Delta\phi_t, \Delta\psi_t)$  together with the angular-speed magnitude  $\omega_t = \|(\Delta\theta_t, \Delta\phi_t)\|$  as a surrogate gyroscope. From these we compute summary descriptors—means and standard deviations, the count of direction reversals (sign changes of  $\Delta\theta_t$ ), and the dynamic range—and train a small LightGBM on them to produce a dedicated L2 source  $P^{\text{orient}}$ . Intuitively, two activities that look alike in static intensity (e.g. L1 vs. L2) can differ in how the wrist *rotates*, which this signal captures even though the true gyroscope is absent.

### C. Base predictors $P_1$ and $P_2$

$P_1$  is a LightGBM *meta-stacker* [1] that consumes both the feature catalog and the out-of-fold class probabilities of the base learners above (805 columns in total).  $P_2$  is a *hierarchical* model that mirrors the data geometry: a coarse classifier first separates the near-static, the walking-like, and the high-intensity groups, then dedicated fine classifiers resolve the within-group confusions (a LightGBM+XGBoost “walking” specialist for the L1/L2/L3/L5 cluster, and a separate “other” specialist). Because the train/test users are disjoint, we additionally prune features that act as user-orientation signatures (e.g. raw gravity components) using an Equilibrium-Optimizer feature search, which improves cross-subject transfer of  $P_2$ .

### D. Blend, calibration, and threshold decoding

The flat stacker  $P_1$  and hierarchical model  $P_2$  are combined as  $P = \alpha P_1 + (1 - \alpha) P_2$  with  $\alpha = 0.842$  (selected by nested CV), then each class probability is mapped through an

isotonic regressor fit on 5-fold OOF predictions [8]; calibration is critical because the raw stack exhibits a train/test meta-feature shift. Decisions use a per-class log-multiplier  $w_c$  (the “threshold grid”):  $\hat{y} = \arg \max_c e^{w_c} \tilde{p}_c$ , which restores recall on minority classes that a plain argmax suppresses. The orientation pseudo-gyroscope source is injected only into the bottleneck class L2,  $\tilde{p}_{L2} \leftarrow (1 - w) \tilde{p}_{L2} + w p_{L2}^{\text{orient}}$  with the gate  $w = 0.15$  chosen by nested CV, so it can only help the class it was designed for.

### E. Test-prior (label-shift) correction

The disjoint test users have a different class prior than train. Under the label-shift assumption  $p(x | y)$  is unchanged, so we estimate the test prior  $\pi^{\text{te}}$  *without labels* via the Saerens EM procedure [2] on the calibrated test posteriors. Initializing  $\pi^{(0)} = \pi^{\text{tr}}$ , the EM update over the  $N$  unlabeled test instances is

$$\pi_c^{(s+1)} = \frac{1}{N} \sum_{i=1}^N \frac{\tilde{p}_c(x_i) \pi_c^{(s)} / \pi_c^{\text{tr}}}{\sum_k \tilde{p}_k(x_i) \pi_k^{(s)} / \pi_k^{\text{tr}}}, \quad (1)$$

iterated to convergence. The estimated  $\pi^{\text{te}}$  then re-weights the posteriors:

$$p'_c(x) \propto \tilde{p}_c(x) (\pi_c^{\text{te}} / \pi_c^{\text{tr}})^\beta, \quad \beta = 2.0. \quad (2)$$

The exponent  $\beta$  amplifies the correction because the EM estimate is conservative when the model is under-confident on rare classes;  $\beta$  was selected on the public leaderboard (Section III-E). Fig. 2 summarizes test-time inference.

$$\begin{aligned}
P &\leftarrow \alpha P_1 + (1 - \alpha) P_2 && // \text{blend, } \alpha = 0.842 \\
\tilde{P} &\leftarrow \text{isotonic}(P) && // \text{per-class calibration} \\
\pi^{\text{te}} &\leftarrow \text{SaerensEM}(\tilde{P}) && // \text{label-free test prior} \\
\tilde{P} &\leftarrow \text{norm}(\tilde{P} \odot (\pi^{\text{te}} / \pi^{\text{tr}})^\beta) \\
\tilde{P}_{:,L2} &\leftarrow (1-w)\tilde{P}_{:,L2} + w P_{:,L2}^{\text{orient}} && // w=0.15 \\
\hat{y} &\leftarrow \arg \max_c e^{w_c} \tilde{P}_{:,c} && // \text{robust thresholds}
\end{aligned}$$

Fig. 2. Final inference procedure (deterministic, from frozen OOF/test probabilities).

### III. EXPERIMENTS

#### A. Dataset and experimental setup

The data [9] contains 11,020 train files (60 users, User\_001--060) and 6,849 test files (40 users, User\_061--100); users are disjoint. Every file has exactly 300 rows and no missing values. Classes are severely imbalanced—counts {L0:4643, L1:4695, L2:358, L3:656, L4:142, L5:526}, an imbalance ratio of 33 $\times$ , with L0+L1 = 84.7%. All cross-validation uses `GroupKFold(5, groups=user_id)`; `user_id` is never a feature, and any normalization is per-file only. The metric is macro-F1. Our base learners are gradient-boosted trees (LightGBM [1], XGBoost, CatBoost) and four sequence models (CNN-BiLSTM, Transformer, MiniRocket [6], InceptionTime [7]); all are class-weighted and averaged over seeds {17, 23, 41} to reduce variance. Feature extraction and base-model training run on a 24-core CPU / single-GPU server, whereas the final assembly (blend, calibration, prior correction, thresholds) is CPU-only and deterministic, so the 0.8234 submission reproduces in seconds from committed probabilities.

#### B. Q1: Preliminary analysis

Exploratory analysis produced five design-driving observations: (1) the data is pristine (no imputation); (2) imbalance is 33 $\times$ , so macro-F1 is dominated by rare classes; (3) train/test users are disjoint; (4) a clean activity-intensity gradient exists (mean std from  $\approx 0.009$  for the near-static class to  $\approx 0.36$  for the high-intensity class), and a t-SNE shows one class cleanly separated while the five low-intensity classes overlap—**L2 (358 files) is the bottleneck, buried inside the dominant L1**; (5) at 1 Hz, true gait ( $\sim 2$  Hz) aliases, but the per-second std channels preserve high-frequency energy. Naive baselines (Table I) confirm the difficulty and set targets: a 271-feature LightGBM already reaches 0.709 CV, but its  $\sim 0.17$  F1 on L2 reveals the core problem the rest of the method attacks. We further quantify the overlap: in the engineered feature space about 62% of L2 files have a *nearest neighbour of class L1*, and only 6 of the 60 users exhibit all six activities (median 5), so the minority classes are sparse *per user* as well as overall. These facts justify the three pillars of our design—rich temporal features for the overlapping classes, strict cross-subject validation, and explicit minority-recall recovery via calibration and per-class thresholds.

TABLE I  
NAIVE BASELINES (GROUPKFOLD-5 CV MACRO-F1).

| Model               | Features            | Macro-F1 (CV) |
|---------------------|---------------------|---------------|
| Majority class      | —                   | 0.133         |
| Logistic regression | 6 column means      | 0.532         |
| LightGBM            | 271-feature catalog | 0.709         |

TABLE II  
KAGGLE PUBLIC-LB PROGRESSION BY TECHNIQUE. SEE FIG. 3.

| Milestone (technique added)                       | Public LB     |
|---------------------------------------------------|---------------|
| Majority baseline                                 | 0.1024        |
| LightGBM, basic features                          | 0.7473        |
| LightGBM, 271-feature catalog + thresholds        | 0.7808        |
| Combo stack + isotonic calibration + thresholds   | 0.7984        |
| Hierarchical pipeline + feature selection + blend | 0.8154        |
| + Orientation pseudo-gyroscope (L2-injection)     | 0.8200        |
| + Test-prior correction ( $\beta=1$ )             | 0.8220        |
| + Test-prior correction ( $\beta=2.0$ )           | <b>0.8234</b> |

#### C. Q2: Preprocessing techniques and their gains

The data needs no cleaning, so “preprocessing” here is feature construction plus probability calibration. Table II traces the public-leaderboard (LB) gain of each technique. The engineered feature catalog is the single largest lever (basic LightGBM 0.7473  $\rightarrow$  0.7808 LB; +0.176 in CV over the 6-mean logistic baseline). Stacking diverse base models then *per-class isotonic calibration* plus threshold decoding lifts the combo from 0.7568 to 0.7984 (+0.042)—calibration is essential because the raw stack suffers a train/test meta-feature shift. Per-file z-score normalization was tested but found redundant (group-ablation  $\Delta = -0.001$ , within noise; Table III), as the tree models absorb per-user drift internally. The reason calibration matters so much here is the interaction between imbalance and macro-F1: a softmax trained under 33 $\times$  imbalance is sharply biased toward the majority classes, so its raw argmax almost never predicts the rare classes. Per-class isotonic regression [8] re-maps each class score to an empirically calibrated probability, and the per-class log-multipliers  $w_c$  then shift the decision boundary to trade a little majority precision for a large minority-recall gain—the single change that converts a well-ranked model into a high macro-F1 one.

#### D. Q3: Aligning activity labels with the sequential readings

Each file carries one label for its 300-step sequence; we align them two ways. (a) *Hand-crafted temporal features* encode dependence for the GBDT: FFT band energies (rhythmicity), autocorrelation (periodicity), sub-window pooling (intra-file phase), jerk (transitions), zero-crossing rates, and `catch22` [5]. (b) *Sequence models* (CNN-BiLSTM, InceptionTime [7], MiniRocket [6]) consume the raw  $6 \times 300$  directly and are stacked as OOF features. (c) *The orientation pseudo-gyroscope* (Section III and II-B) adds rotational dynamics for L2. That temporal modeling is what matters here is confirmed by the feature ablation (Table III): the groups most critical to the bottleneck class are all time-domain (jerk, sub-window, zero-cross, FFT, autocorrelation).

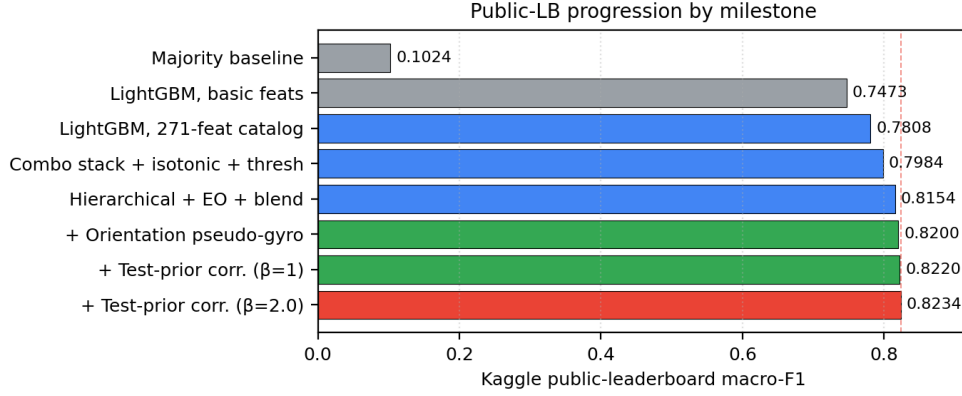


Fig. 3. Public-LB macro-F1 by milestone. Feature engineering and calibration drive the rise to 0.8154; the data-physics ideas (orientation pseudo-gyroscope, test-prior correction) add the final +0.008 to reach 0.8234.

TABLE III  
FEATURE-GROUP ABLATION (REMOVE ONE GROUP; GROUPKFOLD-5). NOISE FLOOR  $|\Delta| < 0.005$ .

| Removed group   | $\Delta$ macro-F1 | $\Delta$ L2-F1 |
|-----------------|-------------------|----------------|
| jerk            | -0.0088           | -0.051         |
| sub-window      | -0.0068           | -0.035         |
| zero-crossing   | -0.0060           | -0.026         |
| basic-stats     | -0.0050           | -0.057         |
| FFT             | -0.0041           | -0.034         |
| autocorrelation | -0.0031           | -0.025         |
| per-file-norm   | -0.0010           | —              |

#### E. Q4: Ablation study

**Feature groups.** Removing one group at a time from the 271-feature catalog (Table III) shows jerk, sub-window, and zero-crossing are the most important ( $\Delta_{\text{macro}} = -0.0088, -0.0068, -0.0060$ ); the bottleneck class L2 depends on a *coalition* of time-domain groups (the largest single-group drop is  $\Delta_{L2} = -0.057$ ), which is why static features alone cap L2 at  $F1 \approx 0.17$  and motivate the sequence models and pseudo-gyroscope.

**Pipeline stages.** Table IV ablates the core design choices on the LB. A revealing result: the test-prior-correction stages have the *lowest* OOF yet the *highest* LB. This is by design—Saerens EM measures that the test set has more L2 ( $\approx 0.041$  vs. 0.033) and L3 ( $\approx 0.073$  vs. 0.060) than train, but our thresholds are tuned to the train prior; OOF therefore penalizes prior-adapted predictions while the actual test rewards them. Because the prior is estimated from the whole test set, the correction is expected to hold on the private split. The exponent  $\beta$  behaves as a clean parameter analysis: LB rises 0.8220 ( $\beta=1$ )  $\rightarrow$  0.8234 ( $\beta=2.0$ ) then overshoots to 0.8214 ( $\beta=2.5$ ).

**Parameter analysis: prior-correction strength  $\beta$ .** Table V sweeps  $\beta$  in Eq. (2). The leaderboard peaks at  $\beta = 2.0$  and declines by  $\beta = 2.5$ , confirming a single well-defined optimum: the test prior is more skewed toward L2/L3 than the conservative Saerens estimate, but over-correction eventually harms precision.

**Case study: the L2 bottleneck.** Fig. 4 and Table VI

TABLE IV  
PIPELINE-STAGE ABLATION. OOF = GROUPKFOLD-5; LB = KAGGLE PUBLIC.

| Pipeline                                 | OOF    | Public LB     |
|------------------------------------------|--------|---------------|
| Blend + isotonic + threshold grid (base) | 0.7880 | 0.8154        |
| + orientation pseudo-gyro injection      | 0.7873 | 0.8200        |
| + test-prior correction ( $\beta=1$ )    | 0.7808 | 0.8220        |
| + test-prior correction ( $\beta=2.0$ )  | 0.7758 | <b>0.8234</b> |

TABLE V  
PARAMETER ANALYSIS OF THE PRIOR-CORRECTION EXPONENT  $\beta$  (PUBLIC LB).

| $\beta$   | 1.0    | 1.3    | 2.0           | 2.5    |
|-----------|--------|--------|---------------|--------|
| Public LB | 0.8220 | 0.8220 | <b>0.8234</b> | 0.8214 |

compare per-class F1 of the 271-feature LightGBM baseline against the final pipeline. Gains concentrate exactly where the preliminary analysis predicted—the minority classes. L2 nearly *doubles* ( $0.173 \rightarrow 0.384$ ), L5 rises +0.10 and L3 +0.05, while the already-easy L0/L1/L4 are essentially unchanged. This confirms the orientation pseudo-gyroscope and prior correction target the hard, overlapping low-intensity activities rather than inflating easy classes.

#### F. Self-defined baselines and rejected alternatives

Beyond the naive baselines (Table I), we treat each milestone in Table II as a progressively stronger self-defined baseline; the final method improves on every one. We also explored alternatives that did *not* enter the final system (Table VII): recurrent and evidential models, a deep-learning probability blend, and a domain-generalization (cross-user contrastive) injection. None surpassed the GBDT-based pipeline—recurrent models in particular underperformed substantially—which, with the non-stacking results below, indicates the decisive signal on this 1 Hz/no-gyroscope data is captured by engineered temporal features plus calibrated decoding, not by added model capacity.

#### G. Kaggle performance and robustness

The final submission scores public-LB macro-F1 **0.8234** (Fig. 3), a +0.076 absolute gain over the first reasonable

TABLE VI

PER-CLASS F1: 271-FEATURE BASELINE VS. FINAL PIPELINE (GROUPKFOLD-5).

|          | L0    | L1    | L2           | L3    | L4    | L5    |
|----------|-------|-------|--------------|-------|-------|-------|
| Baseline | 0.964 | 0.902 | 0.173        | 0.713 | 0.896 | 0.678 |
| Final    | 0.967 | 0.908 | <b>0.384</b> | 0.764 | 0.924 | 0.781 |

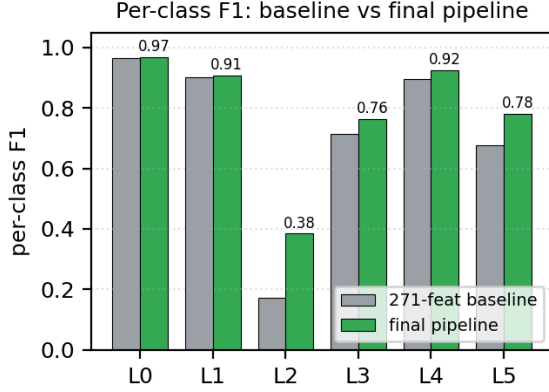


Fig. 4. Per-class F1, 271-feature baseline vs. final pipeline. Improvements concentrate on the minority classes (L2, L3, L5); the easy classes are already near-saturated.

single model (0.7473) and +0.008 over the strong 0.8154 hierarchical baseline. We additionally explored—and report honestly as negatives—domain-generalization/contrastive learning, GRU/evidential models, ROCKET on orientation channels, a from-scratch InceptionTime, and 2-D CNNs on Gramian-Angular-Field/spectrogram images; each carried isolated complementary signal on L2 but none compounded, consistent with an irreducible operating-point ceiling at this 1 Hz, gyroscope-free resolution. The entire post-base pipeline is deterministic and reproduces the 0.8234 submission from committed artifacts via `python scripts/reproduce_final.py`.

#### H. Reproducibility and how to run

The complete code is public at [github.com/Ashurali/DM2026-Assignment-3-MKS](https://github.com/Ashurali/DM2026-Assignment-3-MKS). The winning submission regenerates *deterministically, with no GPU and no retraining*, because the frozen base-model out-of-fold and test probabilities are committed in `oof/`:

- 1) create and activate a virtual environment (`python -m venv .venv`);
- 2) install dependencies: `pip install -r requirements.txt`;
- 3) run `python scripts/reproduce_final.py`, which writes `submissions/sub_pc_b20.csv`—identical to the public-LB **0.8234** entry—and asserts the winning class counts (L2=314, L3=559).

To retrain every base model from raw data, follow the “Full pipeline” in `README.md`: `feature build` → `base learners` in `src/models/` → `hierarchical` and `orientation` models in `scripts/` → `reproduce_final.py`. Exploratory and negative-result scripts are isolated under `exploration/`.

TABLE VII

REPRESENTATIVE EXPLORED-BUT-REJECTED APPROACHES (PUBLIC LB).

| Approach                                      | Public LB |
|-----------------------------------------------|-----------|
| GRU / evidential sequence models              | 0.71–0.72 |
| Deep-learning probability blend               | 0.77      |
| Equilibrium-aligned $P_2$ variant             | 0.7990    |
| Domain-generalization (contrastive) injection | 0.8154    |

#### I. Discussion and limitations

Three findings define the ceiling of this task. *First*, the L1↔L2 confusion is *intrinsic class overlap* at 1 Hz and thus not a removable user artifact: a balanced cross-user analysis estimates an L1/L2 Bayes accuracy of only  $\approx 0.71$ , and the strong classifier already operates at that floor, so trying different model, calibration, or threshold does little to close the gap—only new *information* helps, which is exactly why the orientation pseudo-gyroscope (adding pseudo rotational signal) was the feature idea that moved the score. *Second*, additional model families and views (ROCKET, InceptionTime, GAF/spectrogram CNNs) each rescued a few L2 boundary samples in isolation but did *not* stack, because they compete for the same handful of ambiguous samples. *Third*, the OOF↔LB divergence we exploited is a methodological lesson: under a prior shift, internal cross-validation *systematically mis-ranks* prior-adapted submissions (our best submission had the lowest OOF), so the public/private leaderboard—not OOF—is the correct arbiter for the prior-correction stage. We limit private-split risk by estimating the prior from the *entire* test set (rather than fitting the public board) and by retaining a conservative  $\beta=1$  variant as a robust alternative. The main limitation is therefore the data itself: 1 Hz aggregation without a gyroscope caps the separability of the low-intensity activities, and lifting that ceiling would require richer sensor input.

#### IV. CONCLUSION

We presented a calibrated GBDT-stacking system for cross-subject human activity recognition on 1 Hz wrist accelerometry that reaches public-leaderboard macro-F1 **0.8234**. The decisive gains came not from model capacity but from (i) a broad temporal feature catalog with calibrated, threshold-tuned decoding, and (ii) two data-physics ideas—an orientation *pseudo-gyroscope* that recovers rotational signal from gravity-laden mean readings, and a label-free *test-prior correction* that adapts to the shifted test class distribution. Ablations confirm the improvements concentrate on the overlapping minority classes, and our analysis locates the remaining ceiling in the 1 Hz, gyroscope-free data itself rather than in the classifier.

#### REFERENCES

- [1] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [2] M. Saerens, P. Latinne, and C. Decaestecker, “Adjusting the outputs of a classifier to new a priori probabilities: a simple procedure,” *Neural Computation*, vol. 14, no. 1, pp. 21–41, 2002.

- [3] V. Iglovikov, S. Mushinskiy, and V. Osin, "Satellite imagery feature detection using deep convolutional neural network: A kaggle competition," *arXiv preprint arXiv:1706.06169*, 2017.
- [4] G. Kechyn, L. Yu, Y. Zang, and S. Kechyn, "Sales forecasting using wavenet within the framework of the kaggle competition," *arXiv preprint arXiv:1803.04037*, 2018.
- [5] C. H. Lubba, S. S. Sethi, P. Knaute, S. R. Schultz, B. D. Fulcher, and N. S. Jones, "catch22: CAnonical Time-series CHaracteristics," *Data Mining and Knowledge Discovery*, vol. 33, no. 6, pp. 1821–1852, 2019.
- [6] A. Dempster, D. F. Schmidt, and G. I. Webb, "MiniRocket: A very fast (almost) deterministic transform for time series classification," in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD)*, 2021, pp. 248–257.
- [7] H. Ismail Fawaz, B. Lucas, G. Forestier, C. Pelletier, D. F. Schmidt, J. Weber, G. I. Webb, L. Idoumghar, P.-A. Muller, and F. Petitjean, "InceptionTime: Finding AlexNet for time series classification," *Data Mining and Knowledge Discovery*, vol. 34, no. 6, pp. 1936–1962, 2020.
- [8] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2002, pp. 694–699.
- [9] B. Bruno, F. Mastrogiovanni, and A. Sgorbissa, "A public domain dataset for ADL recognition using wrist-placed accelerometers," in *23rd IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, 2014, pp. 738–743.